

NON-LINEAR SPRING

Consider a chain of N rigid bars of length ℓ , connected at each link by an angular spring of constant k . If each spring has a natural length $x_{i,0} = R\theta_{i,0}$, then the potential energy of the chain is

$$V(|\theta\rangle) = \frac{k}{2} \sum_{i=1}^N R^2 (\theta_i - \theta_{i,0})^2, \quad (1)$$

where θ_i is the current angle between the i -th bar and the $i-1$ -th bar. For convenience, we choose $R = 1$.

Each relative angle θ_i can also be described by the angle that each bar makes with the horizontal axis ϕ_i through the relation $\theta_i = \pi + \phi_i - \phi_{i-1}$. In this basis, the potential energy is

$$V(|\phi\rangle) = \frac{k}{2} \sum_{i=1}^N (\phi_i - \phi_{i-1} - \phi_{i,0} + \phi_{i-1,0})^2. \quad (2)$$

Consider now that the chain is at total length L . Then the system is under the geometric constraint

$$C(|\phi\rangle) = \sum_{i=1}^N \ell \cos \phi_i - L \equiv 0. \quad (3)$$

Thus, if λ_x is the Lagrange multiplier associated with the geometric constraint, the Lagrangian of the system is

$$\mathcal{L}(|\phi\rangle) = -\frac{k}{2} \sum_{i=1}^N (\phi_i - \phi_{i-1} - \phi_{i,0} + \phi_{i-1,0})^2 - \lambda_x \left(\sum_{i=1}^N \ell \cos \phi_i - L \right). \quad (4)$$

The equations of motion are then

$$k(2\phi_i - \phi_{i-1} - \phi_{i+1} - 2\phi_{i,0} + \phi_{i-1,0} + \phi_{i+1,0}) + \lambda_x \ell \sin \phi_i = 0. \quad (5)$$

Consider small perturbations around the natural angles $\phi_i = \phi_{i,0} + \delta\phi_i$. The equations of motion can be rewritten as

$$k(2\delta\phi_i - \delta\phi_{i-1} - \delta\phi_{i+1}) + \lambda_x \ell \sin(\phi_{i,0} + \delta\phi_i) = 0. \quad (6)$$

Expanding the sine and keeping up to first order in $\delta\phi_i$ we have that

$$k(2\delta\phi_i - \delta\phi_{i-1} - \delta\phi_{i+1}) + \lambda_x \ell (\sin \phi_{i,0} + \delta\phi_i \cos \phi_{i,0}) = 0. \quad (7)$$

To get a linear relationship between the Lagrange multiplier λ_x and the deformation $\gamma = \Delta L/L = (L - L_0)/L$, we consider that $\lambda_x \ll 1$ and thus $\lambda_x \delta\phi_i \approx 0$. The equations of motion can be written in vector form as

$$kT|\delta\phi\rangle + \lambda_x \ell |\sin \phi_0\rangle = 0, \quad (8)$$

where T is the tri-diagonal matrix with elements $T_{i,j} = 2\delta_{i,j} - \delta_{i-1,j} - \delta_{i+1,j}$. Thus, if T is invertible, we get

$$|\delta\phi\rangle + \frac{\lambda_x \ell}{k} T^{-1} |\sin\phi_0\rangle = 0. \quad (9)$$

Looking at the geometric constraint in this regime we have that

$$\sum_{i=1}^N \ell \cos(\phi_{i,0} + \delta\phi_i) - L \approx \sum_{i=1}^N [\ell \cos\phi_{i,0} - \ell \sin\phi_{i,0} \delta\phi_i] - L. \quad (10)$$

We know that $\sum_{i=1}^N \ell \cos\phi_{i,0} = L_0$. Additionally, if all natural angles are $\phi_{i,0} = s_i \alpha$ with $s_i = \pm 1$ and $\alpha \in (0, \pi/2)$, we define $|s\rangle = (s_1, \dots, s_N)^T$ and write

$$-\ell \sin\alpha \langle s | \delta\phi \rangle = \Delta L. \quad (11)$$

Substituting Eq. (9) in the above, we get that

$$\sin^2\alpha \frac{\lambda_x \ell^2}{k} \langle s | T^{-1} | s \rangle = \Delta L. \quad (12)$$

Rewriting the above, we get a linear relation between λ_x and γ ,

$$\lambda_x = \frac{L}{\ell^2} \frac{k}{\sin^2\alpha} (\langle s | T^{-1} | s \rangle)^{-1} \gamma. \quad (13)$$

igna: Con Claudio nos preguntabamos como escala $(\langle s | T^{-1} | s \rangle)^{-1}$ con N , o L . Podemos sacar un scaling, al menos numericamente?